

7jfradj5s

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```
[1]: import cv2
      from PIL import Image
      import numpy as np
      import matplotlib.pyplot as plt
      import matplotlib.cm as cm
```

```
[2]: # Load image using OpenCV (automatically stored as a NumPy array)
      img = cv2.imread('Images/cameraman.tif')

      # Display the image using Matplotlib
      plt.imshow(img)
      plt.title("Loaded via OpenCV")
      plt.axis('off') # Hide axis coordinates
      plt.show()
```

Loaded via OpenCV



```
[3]: # Load image using PIL (stored as a PIL Image object)
img2 = Image.open('Images/lena_gray_256.tif')

# Display the image with a grayscale colormap
plt.imshow(img2, cmap=cm.Greys_r)
plt.title("Loaded via PIL (Grayscale)")
plt.axis('off')
plt.show()
```

Loaded via PIL (Grayscale)



```
[4]: # Save the OpenCV image to disk
cv2.imwrite('new_image.jpg', img)

# Save the PIL image to disk
img2.save('new_image2.jpg')

print("Both images have been saved successfully.")
```

Both images have been saved successfully.

```
[5]: # Check shape and print pixel values for the OpenCV image
print("=== OpenCV Image ===")
print("Dimensions (Shape):", img.shape)
print("Pixel Array:\n", img)

# Convert PIL image to a NumPy array, then inspect shape and values
img_array = np.array(img2)
print("\n=== PIL Image (Converted to NumPy) ===")
print("Dimensions (Shape):", img_array.shape)
print("Pixel Array:\n", img_array)
```

```
=== OpenCV Image ===
Dimensions (Shape): (512, 512, 3)
Pixel Array:
```

```
[[[156 156 156]
  [157 157 157]
  [160 160 160]
  ...
  [152 152 152]
  [152 152 152]
  [152 152 152]]]
```

```
[[[156 156 156]
  [157 157 157]
  [159 159 159]
  ...
  [152 152 152]
  [152 152 152]
  [152 152 152]]]
```

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[[[158 158 158]
  [157 157 157]
  [156 156 156]
  ...
  [152 152 152]
  [152 152 152]
  [152 152 152]]]
```

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...
```

```
[[[121 121 121]
  [123 123 123]
  [126 126 126]
  ...
  [121 121 121]
  [113 113 113]
  [111 111 111]]]
```

```

[[121 121 121]
 [123 123 123]
 [126 126 126]
 ...
 [121 121 121]
 [113 113 113]
 [111 111 111]]

[[121 121 121]
 [123 123 123]
 [126 126 126]
 ...
 [121 121 121]
 [113 113 113]
 [111 111 111]]]

```

=== PIL Image (Converted to NumPy) ===

Dimensions (Shape): (256, 256)

Pixel Array:

```

[[162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 ...
 [ 48  57  54 ...  75  97  88]
 [ 46  50  52 ...  89 100  98]
 [ 44  55  54 ...  97 100 105]]

```

```

[6]: # Inspect data type, minimum, and maximum pixel values
print("Data Type:", img_array.dtype)
print("Min Pixel Value:", np.min(img_array))
print("Max Pixel Value:", np.max(img_array))

# Crop a region of interest using array slicing
cropped_img = img_array[50:200, 50:200]

# Visualize the cropped region
plt.imshow(cropped_img, cmap='gray')
plt.title("Cropped Region")
plt.axis('off')
plt.show()

```

Data Type: uint8

Min Pixel Value: 26

Max Pixel Value: 242

Cropped Region

